Anatomy of Flowering Plants

The external morphology of larger living organisms - plants and animals alike - shows clear structural similarities and variations; a study of their **internal** structure reveals similarities and differences too. This chapter deals with the internal structure and functional organisation of higher plants.

- **Anatomy** - the study of the **internal structure of plants**.

Plants have **cells as the basic unit**; cells are organised into **tissues**, and in turn the tissues are organised into **organs**.

- **Different organs** in a plant show differences in their internal structure.
- Within **angiosperms**, the **monocots and dicots** are also seen to be **anatomically different**.
- Internal structures also show **adaptations to diverse environments**.

6.1 The Tissue System

Tissues were earlier discussed on the basis of the **types of cells** present. Tissues also vary depending on their **location** in the plant body, and their **structure and function are dependent on that location**.

The plant tissues are broadly classified into **meristematic** (apical, lateral and intercalary) and **permanent** (simple and complex) tissues. **Assimilation of food and its storage, transportation of water, minerals and photosynthates, and mechanical support** are the main functions of tissues.

On the basis of their structure and location, there are **three types of tissue systems**:

- the **epidermal tissue system**,
- the **ground or fundamental tissue system**, and
- the **vascular or conducting tissue system**.

6.1.1 Epidermal Tissue System

The epidermal tissue system forms the **outer-most covering of the whole plant body** and comprises **epidermal cells, stomata** and the epidermal appendages - the **trichomes and hairs**.

The **epidermis** is the outermost layer of the primary plant body. It is made up of **elongated, compactly arranged cells**, which form a **continuous layer**.

- Epidermis is **usually single-layered**.
- Epidermal cells are **parenchymatous** with a small amount of cytoplasm lining the cell wall and a **large vacuole**.
- The outside of the epidermis is often covered with a **waxy thick layer called the cuticle**, which prevents the loss of water. **Cuticle is absent in roots**.

Epidermal appendages. The cells of epidermis bear a number of hairs.

- **Root hairs** are **unicellular** elongations of the epidermal cells and help absorb **water and minerals from the soil**.
- On the stem the epidermal hairs are called **trichomes**. The trichomes in the shoot system are **usually multicellular**.
- They **may be branched or unbranched** and **soft or stiff**; they may even be **secretory**.
- Trichomes help in **preventing water loss due to transpiration**.

- **Stomata -** structures present in the epidermis of **leaves** that regulate the process of **transpiration and gaseous exchange**.

- Each stoma is composed of **two bean-shaped cells known as guard cells** which enclose the **stomatal pore**. **In grasses, the guard cells are dumb-bell shaped**.
- The **outer walls** of guard cells (away from the stomatal pore) are **thin** and the **inner walls** (towards the stomatal pore) are **highly thickened**.
- Guard cells possess **chloroplasts** and regulate the **opening and closing of stomata**.
- **Sometimes**, a few epidermal cells in the vicinity of the guard cells become specialised in shape and size and are known as **subsidiary cells**.

- **Stomatal apparatus -** the **stomatal aperture, guard cells and the surrounding subsidiary cells** together.

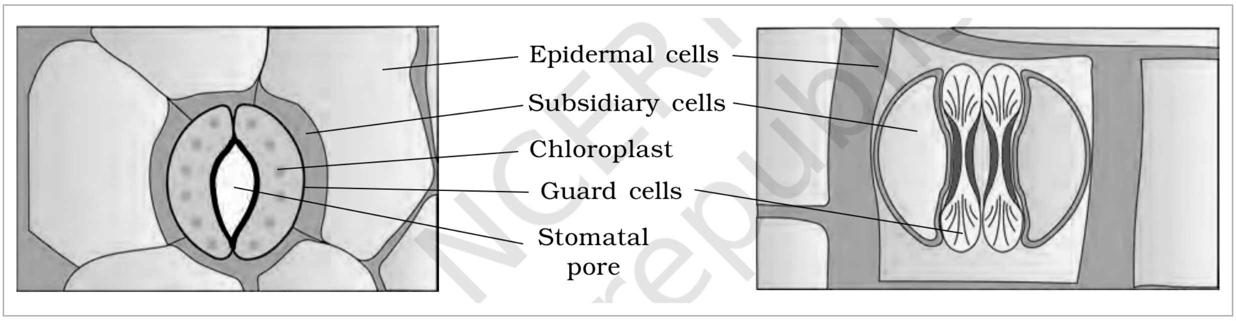

Figure 6.1 Diagrammatic representation: (a) stomata with bean-shaped guard cells (b) stomata with dumb-bell shaped guard cell

ⓘ ***[NOTE]*** *Figure 6.1 labels (verbatim):* ***Epidermal cells****;* ***Subsidiary cells****;* ***Chloroplast****;* ***Guard cells****;* ***Stomatal pore****.*

6.1.2 The Ground Tissue System

All tissues except epidermis and vascular bundles constitute the ground tissue. The ground tissue system **forms the main bulk of the plant**.

- It consists of **simple tissues** such as **parenchyma, collenchyma and sclerenchyma**.
- **Parenchymatous cells** are **usually present in cortex, pericycle, pith and medullary rays**, in the primary stems and roots.
- In **leaves**, the ground tissue consists of thin-walled, chloroplast-containing cells and is called **mesophyll**.
- The ground tissue is divided into **three zones - cortex, pericycle and pith**.

6.1.3 The Vascular Tissue System

The vascular system consists of **complex tissues, the phloem and the xylem**. The **xylem and phloem together constitute vascular bundles**. The vascular bundles form the **conducting tissue** and **translocate water, minerals and food material**.

Open vs closed vascular bundles (basis: presence of cambium):

- **Open -** in **dicotyledonous stems, cambium is present between phloem and xylem**. Because of this cambium these bundles can form **secondary xylem and phloem**, and are hence called **open**.
- **Closed -** in **monocotyledons, the vascular bundles have no cambium**. Since they do **not form secondary tissues** they are referred to as **closed**.

Radial vs conjoint (basis: relative position of xylem and phloem):

- **Radial -** when xylem and phloem are arranged in an **alternate manner along different radii**, such as **in roots**.
- **Conjoint -** when xylem and phloem are **jointly situated along the same radius**. Such bundles are **common in stems and leaves**, and **usually have the phloem located only on the outer side of the xylem**.

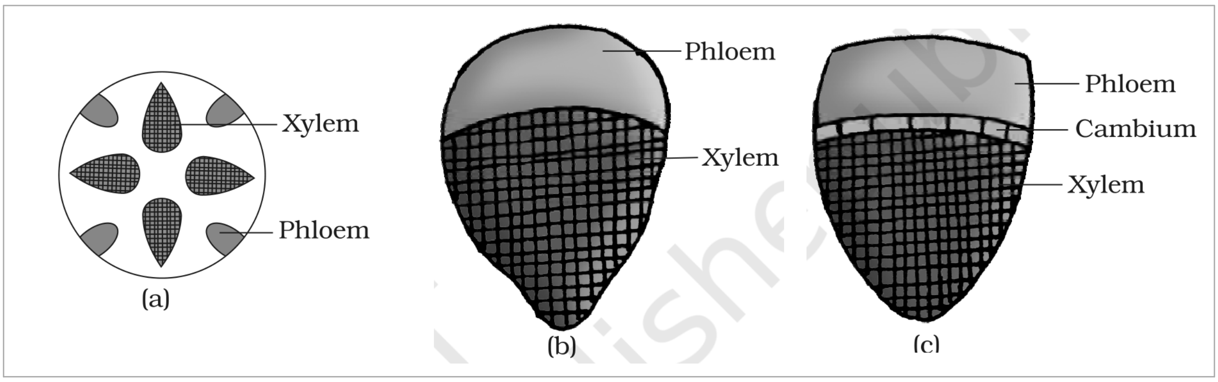

Figure 6.2 Various types of vascular bundles: (a) radial (b) conjoint closed (c) conjoint open

ⓘ ***[NOTE]*** *Figure 6.2 labels (verbatim):* ***Xylem****;* ***Phloem****;* ***Cambium****.*

6.2 Anatomy of Dicotyledonous and Monocotyledonous Plants

For a better understanding of tissue organisation of **roots, stems and leaves**, it is convenient to study the **transverse sections of the mature zones** of these organs. **Secondary growth occurs in most of the**

dicotyledonous roots and stems.

6.2.1 Dicotyledonous Root

Figure 6.3 (a) shows the transverse section of the **sunflower root**. From outside in, the internal tissue organisation is as follows:

- **Epiblema -** the outermost layer. Many of its cells protrude as **unicellular root hairs**.
- **Cortex -** several layers of **thin-walled parenchyma cells with intercellular spaces**.
- **Endodermis -** the innermost layer of the cortex; a **single layer of barrel-shaped cells without any intercellular spaces**. Its **tangential and radial walls** have a deposition of the water-impermeable waxy material **suberin** in the form of **casparian strips**.
- **Pericycle -** a few layers of **thick-walled parenchymatous cells** next to the endodermis. **Initiation of lateral roots and of vascular cambium during secondary growth** takes place in these cells.
- **Conjunctive tissue -** the parenchymatous cells that lie **between the xylem and the phloem**.
- **Pith - small or inconspicuous** in the dicot root.

Vascular tissue and stele.

- There are **usually two to four (2-4) xylem and phloem patches**.
- Later, a **cambium ring develops between the xylem and phloem**.

- **Stele -** all tissues on the inner side of the endodermis, i.e. **pericycle, vascular bundles and pith**, together.

6.2.2 Monocotyledonous Root

The anatomy of the monocot root is **similar to the dicot root in many respects**. It has **epidermis, cortex, endodermis, pericycle, vascular bundles and pith**.

- As compared to the dicot root (which has fewer xylem bundles), there are **usually more than six (polyarch) xylem bundles** in the monocot root.
- **Pith is large and well developed.**
- **Monocotyledonous roots do not undergo any secondary growth.**

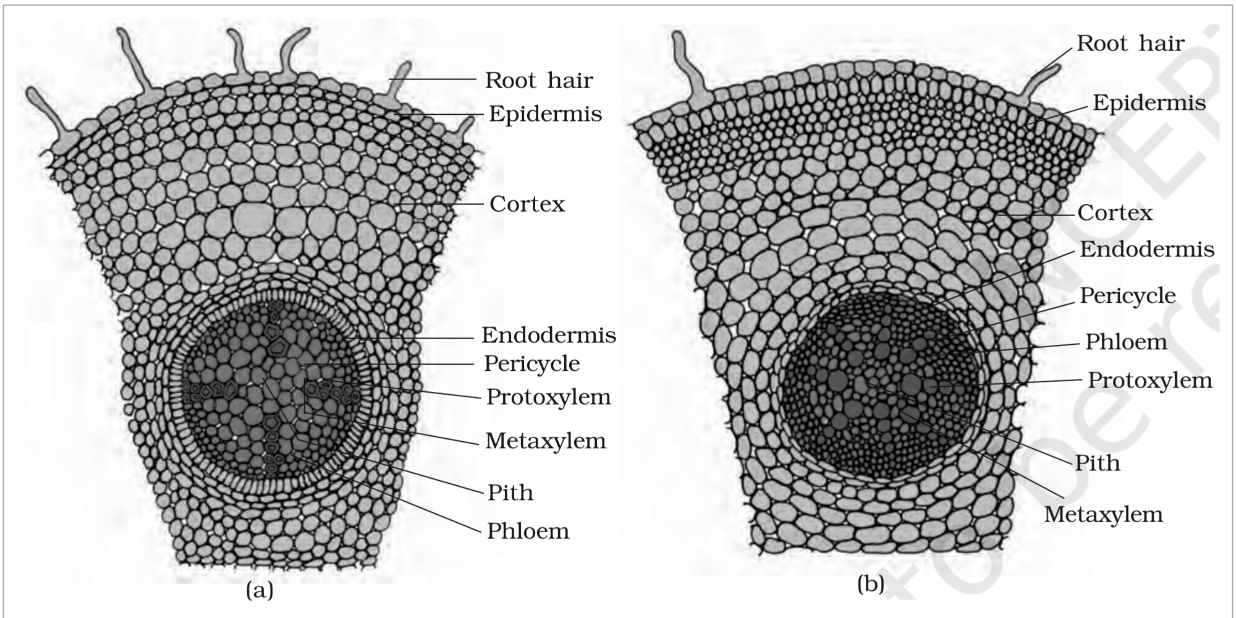

Figure 6.3 T.S.: (a) Dicot root (Primary) (b) Monocot root

> ⓘ ***[NOTE]** Figure 6.3 labels (verbatim): **Root hair**; **Epidermis**; **Cortex**; **Endodermis**; **Pericycle**; **Protoxylem**; **Metaxylem**; **Pith**; **Phloem**.*

6.2.3 Dicotyledonous Stem

The transverse section of a typical young dicotyledonous stem shows the **epidermis** as the outermost protective layer. Covered with a thin layer of **cuticle**, it may bear **trichomes and a few stomata**.

Cortex. The cells arranged in **multiple layers between epidermis and pericycle** constitute the cortex, which consists of **three sub-zones**:

- **Hypodermis** (outer) **-** a few layers of **collenchymatous cells** just below the epidermis, which provide **mechanical strength** to the young stem.
- **Cortical layers below the hypodermis - rounded, thin-walled parenchymatous cells** with conspicuous intercellular spaces.
- **Endodermis** (innermost layer of the cortex) **-** its cells are **rich in starch grains**, so the layer is also called the **starch sheath**.
- **Pericycle -** present on the inner side of the endodermis and above the phloem, in the form of **semi-lunar patches of sclerenchyma**.
- **Medullary rays -** a few layers of radially placed parenchymatous cells **in between the vascular bundles**.
- **Vascular bundles -** a large number arranged **in a ring**; this 'ring' arrangement is a **characteristic of the dicot stem**. Each bundle is **conjoint, open, and with endarch protoxylem**.
- **Pith -** a large number of rounded parenchymatous cells with large intercellular spaces occupying the **central portion** of the stem.

6.2.4 Monocotyledonous Stem

The monocot stem has a **sclerenchymatous hypodermis**, a large number of **scattered vascular bundles** (each surrounded by a **sclerenchymatous bundle sheath**), and a **large, conspicuous parenchymatous ground tissue**.

- Vascular bundles are **conjoint and closed**.
- **Peripheral vascular bundles are generally smaller** than the centrally located ones.
- The **phloem parenchyma is absent**, and **water-containing cavities** are present within the vascular bundles.

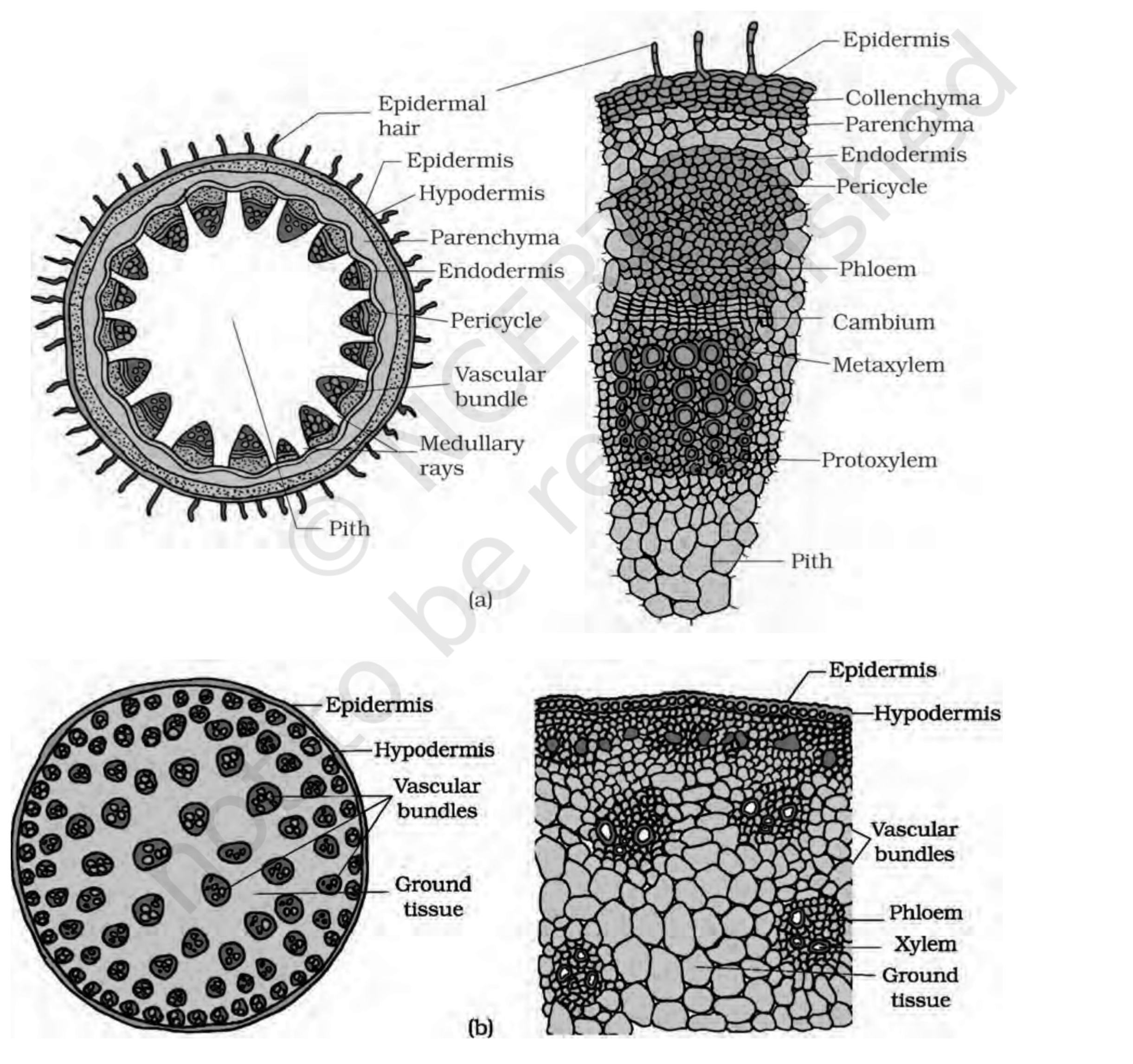

Figure 6.4 T.S. of stem: (a) Dicot (b) Monocot

ⓘ ***[NOTE]*** *Figure 6.4 labels (verbatim):* ***Epidermal hair****;* ***Epidermis****;* ***Hypodermis****;* ***Parenchyma****;* ***Endodermis****;* ***Pericycle****;* ***Vascular bundle****;* ***Medullary rays****;* ***Pith****;* ***Collenchyma****;* ***Phloem****;* ***Cambium****;* ***Metaxylem****;* ***Protoxylem****;* ***Xylem****;* ***Vascular bundles****;* ***Ground tissue****.*

6.2.5 Dorsiventral (Dicotyledonous) Leaf

The vertical section of a dorsiventral leaf through the lamina shows **three main parts: epidermis, mesophyll and vascular system**.

Epidermis.

- It covers both the **upper surface (adaxial epidermis)** and the **lower surface (abaxial epidermis)** of the leaf and has a **conspicuous cuticle**.
- The **abaxial epidermis generally bears more stomata** than the adaxial epidermis; the **latter (adaxial) may even lack stomata**.

Mesophyll.

- The tissue **between the upper and the lower epidermis** is the mesophyll. It possesses **chloroplasts**, carries out **photosynthesis**, and is made up of **parenchyma**.
- It has **two types of cells -** the **palisade parenchyma** and the **spongy parenchyma**.
- The **adaxially placed palisade parenchyma** is made up of **elongated cells arranged vertically and parallel** to each other.

- The **oval or round, loosely arranged spongy parenchyma** lies **below the palisade cells** and extends to the lower epidermis, with **numerous large spaces and air cavities** between the cells.

Vascular system.

- It includes **vascular bundles**, seen in the **veins and the midrib**. The **size of the vascular bundles depends on the size of the veins**, and the veins **vary in thickness in the reticulate venation** of the dicot leaves.
- The vascular bundles are surrounded by a layer of **thick-walled bundle sheath cells**.
- Note the **position of the xylem** within the bundle (Figure 6.5 a): the **xylem lies towards the adaxial (upper) side** and the **phloem towards the abaxial (lower) side**.

6.2.6 Isobilateral (Monocotyledonous) Leaf

The anatomy of the isobilateral leaf is **similar to that of the dorsiventral leaf in many ways**, with the following characteristic differences:

- **Stomata are present on both surfaces** of the epidermis; and the **mesophyll is not differentiated into palisade and spongy parenchyma**.
- **Bulliform cells -** in **grasses**, certain adaxial epidermal cells along the veins modify into **large, empty, colourless cells**.
- When the bulliform cells have **absorbed water and are turgid**, the **leaf surface is exposed**.
- When they are **flaccid due to water stress**, they make the **leaves curl inwards to minimise water loss**.
- The **parallel venation** in monocot leaves is reflected in the **near similar sizes of vascular bundles (except in main veins)**, as seen in vertical sections.

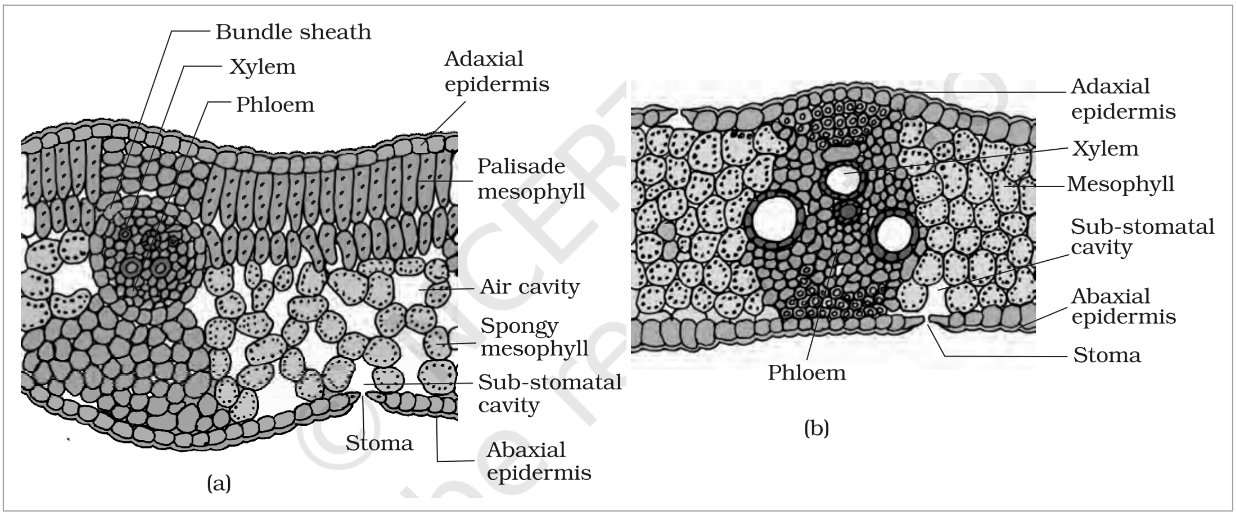

Figure 6.5 T.S. of leaf: (a) Dicot (b) Monocot

> ⓘ ***[NOTE]*** *Figure 6.5 labels (verbatim):* ***Bundle sheath****;* ***Xylem****;* ***Phloem****;* ***Adaxial epidermis****;* ***Palisade mesophyll****;* ***Air cavity****;* ***Spongy mesophyll****;* ***Sub-stomatal cavity****;* ***Stoma****;* ***Abaxial epidermis****;* ***Mesophyll****.*

Recap Quick Recap

- Anatomically, a plant is made of different kinds of **tissues**, broadly **meristematic** (apical, lateral, intercalary) and **permanent** (simple, complex). Their main functions are **assimilation and storage of food, transport of water, minerals and photosynthates, and mechanical support**.
- There are **three tissue systems - epidermal, ground and vascular**. The epidermal system is made of **epidermal cells, stomata and epidermal appendages**.
- The **ground tissue system forms the main bulk** of the plant and is divided into **three zones - cortex, pericycle and pith**.
- The **vascular tissue system** is formed by **xylem and phloem**; on the basis of **presence of cambium and location of xylem and phloem** the vascular bundles are of different types. They form the **conducting tissue and translocate water, minerals and food material**.
- **Monocots and dicots differ** in the **type, number and location of vascular bundles**, and **secondary growth occurs in most dicot roots and stems**.

Q Terms Used in the Exercises

Q6. How is the study of plant anatomy useful to us?

- **Anatomy is the study of the internal structure and functional organisation of plants**, so it lets us understand how a plant is built from cells to tissues to organs.
- It lets us **distinguish monocots from dicots anatomically -** within angiosperms the two are internally different **-** which is how a stem or root is identified from a transverse section.
- It reveals the **adaptations of internal structures to diverse environments**, e.g. bulliform cells that curl a grass leaf to reduce water loss.